Class11_AlphaFold

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Interpreting Results 1

##Querying the AlphaFold database

Search the sequence in AlphaFold database

My top hits: Gag-Pol polyprotein AF-0000000365763110-v1

Interpreting Results

```
img <- png::readPNG("AF-A0A6L8LSL6-F1-MODEL_V6.PDB-AF-0000000365763110-MODEL_V1.PDB.png")

plot(0:1, 0:1, type = "n", xlab = "", ylab = "", axes = FALSE)
rasterImage(img, 0, 0, 1, 1)
```

Insert the AlhaFlod results into R

```
library(bio3d)
results_dir <- "HIVprdimer_23119"
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)

# File names for all PDB models
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)

# Print our PDB file names
basename(pdb_files)
```

```
[1] "HIVprdimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_000.pdb"
[2] "HIVprdimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_4_seed_000.pdb"
[3] "HIVprdimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_1_seed_000.pdb"
[4] "HIVprdimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_5_seed_000.pdb"
[5] "HIVprdimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```r
# Read all data from Models
#  and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
HIVprdimer_23119/HIVprdimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_000
HIVprdimer_23119/HIVprdimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_4_seed_000
HIVprdimer_23119/HIVprdimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_1_seed_000
HIVprdimer_23119/HIVprdimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_5_seed_000
HIVprdimer_23119/HIVprdimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000
.....

Extracting sequences

pdb/seq: 1   name: HIVprdimer_23119/HIVprdimer_23119_unrelaxed_rank_001_alphafold2_multimer_v
pdb/seq: 2   name: HIVprdimer_23119/HIVprdimer_23119_unrelaxed_rank_002_alphafold2_multimer_v
pdb/seq: 3   name: HIVprdimer_23119/HIVprdimer_23119_unrelaxed_rank_003_alphafold2_multimer_v
pdb/seq: 4   name: HIVprdimer_23119/HIVprdimer_23119_unrelaxed_rank_004_alphafold2_multimer_v
pdb/seq: 5   name: HIVprdimer_23119/HIVprdimer_23119_unrelaxed_rank_005_alphafold2_multimer_v
```

```r
pdbs
```

```
                              1        .         .         .         .         50
[Truncated_Name:1]HIVprdimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]HIVprdimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]HIVprdimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]HIVprdimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]HIVprdimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                              **************************************************
                              1        .         .         .         .         50

                             51        .         .         .         .         100
[Truncated_Name:1]HIVprdimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]HIVprdimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]HIVprdimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]HIVprdimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]HIVprdimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                              **************************************************
                             51        .         .         .         .         100

                            101        .         .         .         .         150
```

```
[Truncated_Name:1]HIVprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]HIVprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]HIVprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]HIVprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]HIVprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                               **************************************************
                             101        .         .         .         .         150

                             151        .         .         .         .       198
[Truncated_Name:1]HIVprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]HIVprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]HIVprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]HIVprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]HIVprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                               **************************************************
                             151        .         .         .         .       198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

RMSD is a standard measure of structural distance between coordinate sets. We can use the rmsd() function to calculate the RMSD between all pairs models.

```
rd <- rmsd(pdbs, fit=T)
```

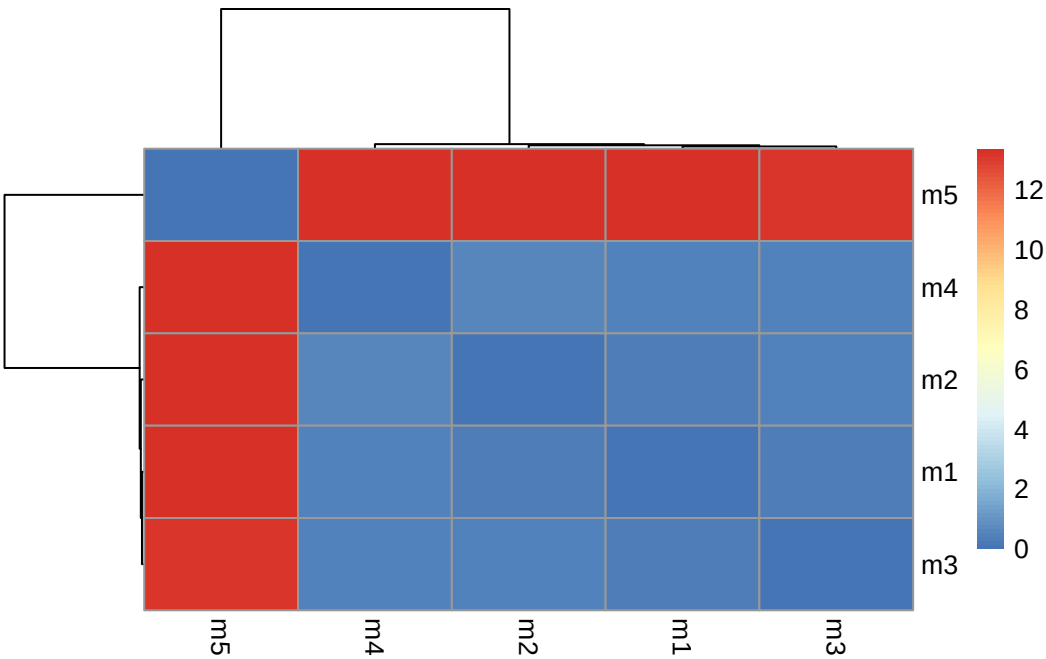
```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```
range(rd)
```

```
[1]  0.000 13.376
```

```r
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```

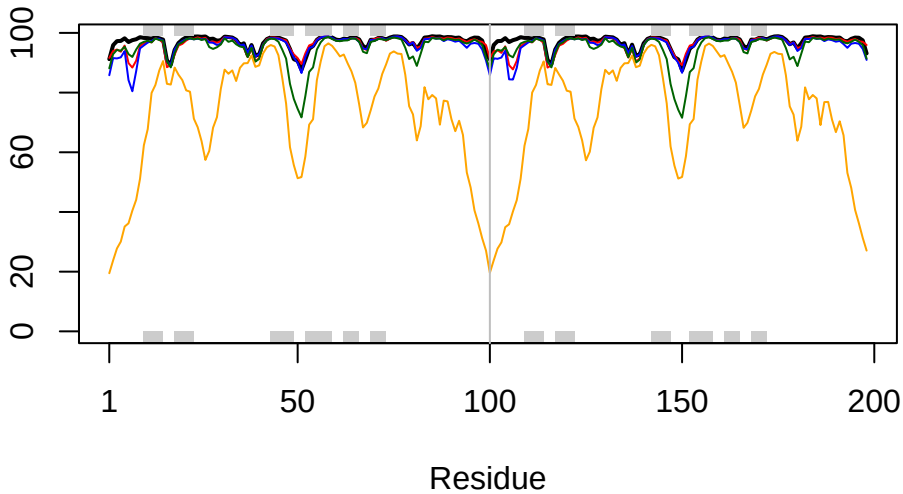


Now lets plot the pLDDT values across all models. Recall that this information is in the B-factor column of each model and that this is stored in our aligned `pdbs` object as `pdbs$b` with a row per structure/model.

```r
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



We can improve the superposition/fitting of our models by finding the most consistent "rigid core" common across all the models. For this we will use the `core.find()` function:

```
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 58.625
core size 196 of 198  vol = 52.96
core size 195 of 198  vol = 49.614
core size 194 of 198  vol = 46.269
core size 193 of 198  vol = 44.802
core size 192 of 198  vol = 43.508
core size 191 of 198  vol = 42.246
core size 190 of 198  vol = 40.982
core size 189 of 198  vol = 39.788
core size 188 of 198  vol = 38.576
core size 187 of 198  vol = 37.347
core size 186 of 198  vol = 36.342
core size 185 of 198  vol = 35.226
core size 184 of 198  vol = 34.286
core size 183 of 198  vol = 33.532
core size 182 of 198  vol = 32.75
core size 181 of 198  vol = 32.195
core size 180 of 198  vol = 31.494
```

```
core size 179 of 198  vol = 31.013
core size 178 of 198  vol = 30.448
core size 177 of 198  vol = 29.837
core size 176 of 198  vol = 29.536
core size 175 of 198  vol = 29.483
core size 174 of 198  vol = 29.556
core size 173 of 198  vol = 29.367
core size 172 of 198  vol = 29.366
core size 171 of 198  vol = 29.271
core size 170 of 198  vol = 29.272
core size 169 of 198  vol = 29.032
core size 168 of 198  vol = 28.853
core size 167 of 198  vol = 28.495
core size 166 of 198  vol = 28.045
core size 165 of 198  vol = 27.445
core size 164 of 198  vol = 26.7
core size 163 of 198  vol = 26.022
core size 162 of 198  vol = 25.412
core size 161 of 198  vol = 24.67
core size 160 of 198  vol = 24.117
core size 159 of 198  vol = 23.679
core size 158 of 198  vol = 23.089
core size 157 of 198  vol = 22.457
core size 156 of 198  vol = 22.038
core size 155 of 198  vol = 21.617
core size 154 of 198  vol = 21.215
core size 153 of 198  vol = 20.757
core size 152 of 198  vol = 19.861
core size 151 of 198  vol = 19.611
core size 150 of 198  vol = 19.177
core size 149 of 198  vol = 18.875
core size 148 of 198  vol = 18.447
core size 147 of 198  vol = 17.829
core size 146 of 198  vol = 17.441
core size 145 of 198  vol = 17.063
core size 144 of 198  vol = 16.662
core size 143 of 198  vol = 16.4
core size 142 of 198  vol = 15.791
core size 141 of 198  vol = 15.113
core size 140 of 198  vol = 14.581
core size 139 of 198  vol = 14.037
core size 138 of 198  vol = 13.63
core size 137 of 198  vol = 12.821
```

```
core size 136 of 198  vol = 12.496
core size 135 of 198  vol = 12.232
core size 134 of 198  vol = 11.95
core size 133 of 198  vol = 11.705
core size 132 of 198  vol = 11.36
core size 131 of 198  vol = 10.983
core size 130 of 198  vol = 10.545
core size 129 of 198  vol = 9.983
core size 128 of 198  vol = 9.562
core size 127 of 198  vol = 9.27
core size 126 of 198  vol = 8.703
core size 125 of 198  vol = 8.091
core size 124 of 198  vol = 7.644
core size 123 of 198  vol = 7.291
core size 122 of 198  vol = 6.936
core size 121 of 198  vol = 6.575
core size 120 of 198  vol = 6.114
core size 119 of 198  vol = 5.689
core size 118 of 198  vol = 5.281
core size 117 of 198  vol = 4.892
core size 116 of 198  vol = 4.582
core size 115 of 198  vol = 4.307
core size 114 of 198  vol = 4.056
core size 113 of 198  vol = 3.72
core size 112 of 198  vol = 3.561
core size 111 of 198  vol = 3.409
core size 110 of 198  vol = 3.276
core size 109 of 198  vol = 3.035
core size 108 of 198  vol = 2.97
core size 107 of 198  vol = 2.859
core size 106 of 198  vol = 2.784
core size 105 of 198  vol = 2.484
core size 104 of 198  vol = 2.279
core size 103 of 198  vol = 2.203
core size 102 of 198  vol = 2.111
core size 101 of 198  vol = 2.002
core size 100 of 198  vol = 1.905
core size 99 of 198  vol = 1.788
core size 98 of 198  vol = 1.673
core size 97 of 198  vol = 1.526
core size 96 of 198  vol = 1.403
core size 95 of 198  vol = 1.165
core size 94 of 198  vol = 0.987
```

```
 core size 93 of 198  vol = 0.803
 core size 92 of 198  vol = 0.727
 core size 91 of 198  vol = 0.587
 core size 90 of 198  vol = 0.525
 core size 89 of 198  vol = 0.396
 FINISHED: Min vol ( 0.5 ) reached
```

```
core.inds <- print(core, vol=0.5)
```

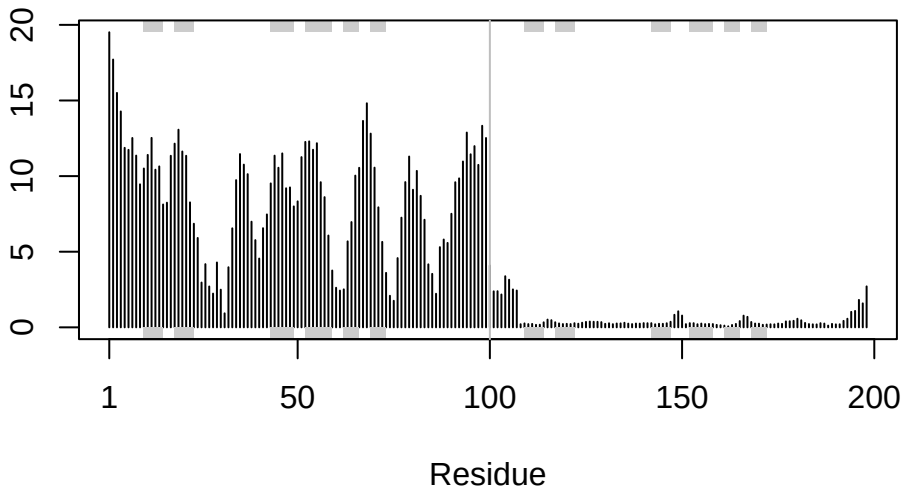
```
# 90 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  78     27
3    80  97     18
```

```
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

Now we can examine the RMSF between positions of the structure. RMSF is an often used measure of conformational variance along the structure:

```
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



##Predicted Alignment Error for domain

Independent of the 3D structure, AlphaFold produces an output called **Predicted Aligned Error (PAE)**. This is detailed in the JSON format result files, one for each model structure.

```
library(jsonlite)

# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)

pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)
pae2 <- read_json(pae_files[2],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```r
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 91.19 95.88 97.25 97.31 98.19 96.94
```

The maximum PAE values are useful for ranking models. The lower the PAE score the better.

> Q. How about the other models, what are thir max PAE scores?

```r
pae1$max_pae
```

```
[1] 13.78125
```

```r
pae5$max_pae
```

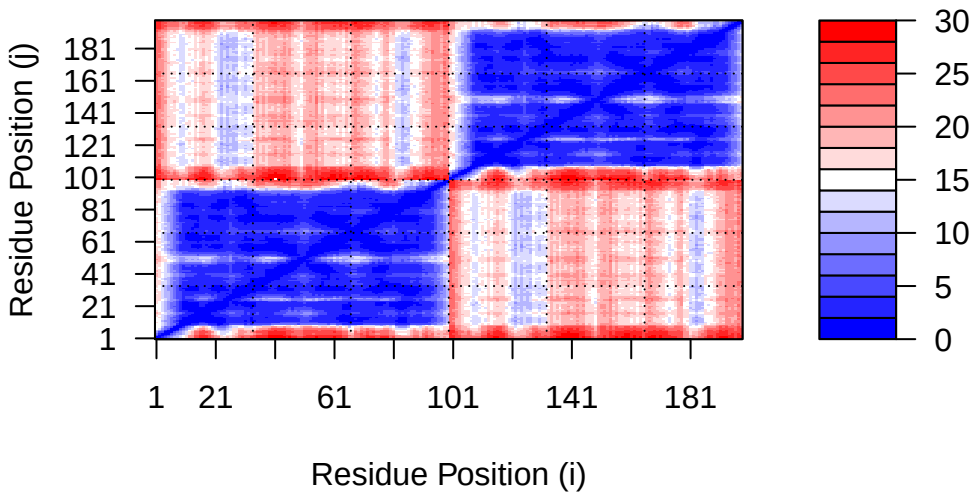
```
[1] 30.01562
```

```r
pae2$max_pae
```

```
[1] 12.98438
```

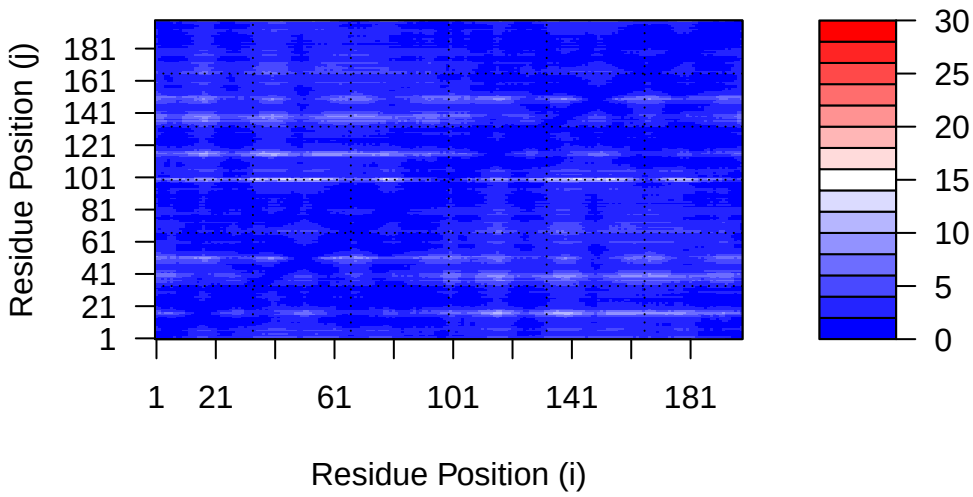
We can plot the N by N (where N is the number of residues) PAE scores with ggplot or with functions from the Bio3D package:

```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



We should really plot all of these using the same z range. Here is the model 1 plot again but this time using the same data range as the plot for model 5:

```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



##Residue conservation from alignment file

Use code to analyze residue conservation from a multiple sequence alignment

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "HIVprdimer_23119/HIVprdimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

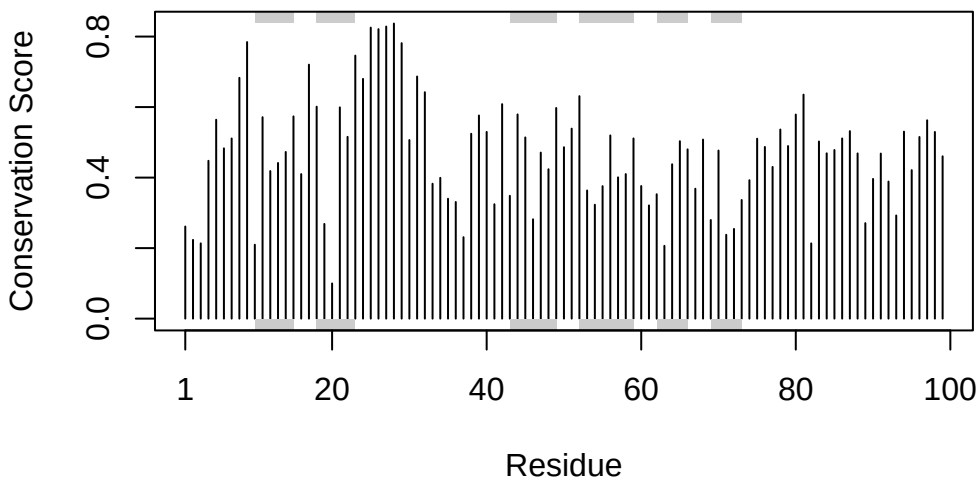
```
dim(aln$ali)
```

```
[1] 5397  132
```

```r
sim <- conserv(aln)
```

Use code tp plot the conservation scores for the first 99 residues and overlays them with secondary structure information from chain A of a PDB structure.

```r
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



Note the conserved Active Site residues D25, T26, G27, A28. These positions will stand out if we generate a consensus sequence with a high cutoff value:

```r
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

For a final visualization of these functionally important sites we can map this conservation score to the Occupancy column of a PDB file for viewing in molecular viewer programs such as Mol*, PyMol, VMD, chimera etc.

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

```
knitr::include_graphics("M1_CONSERV.PDB.png")
```